

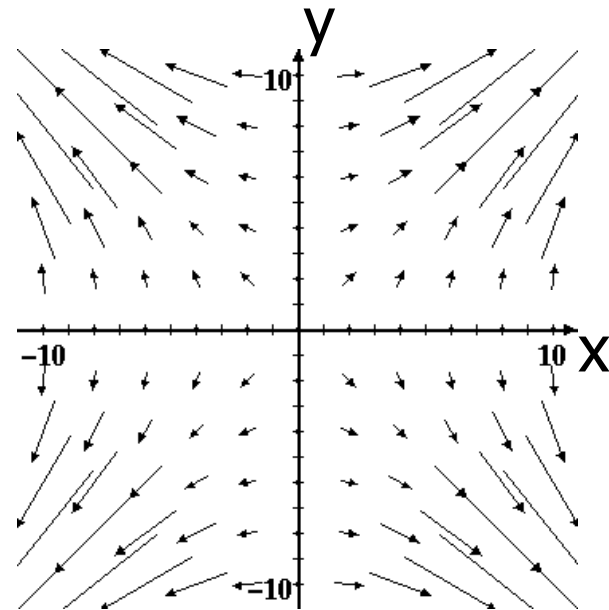
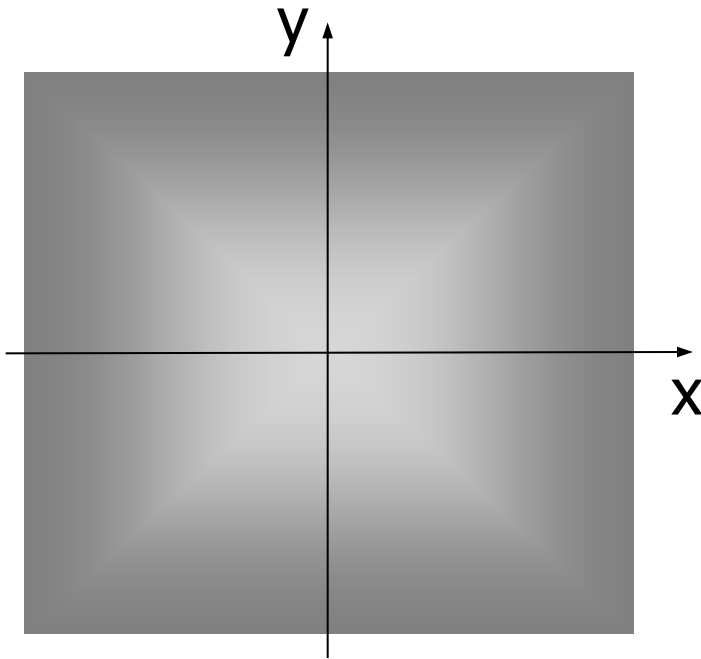
第1章 矢量场

1.1 矢量及其矢量场

1. 什么是标量、矢量（向量）

标量： 只有大小

矢量： 既有大小，又有方向



2. 矢量表示方法

矢量几何表示：有向线段



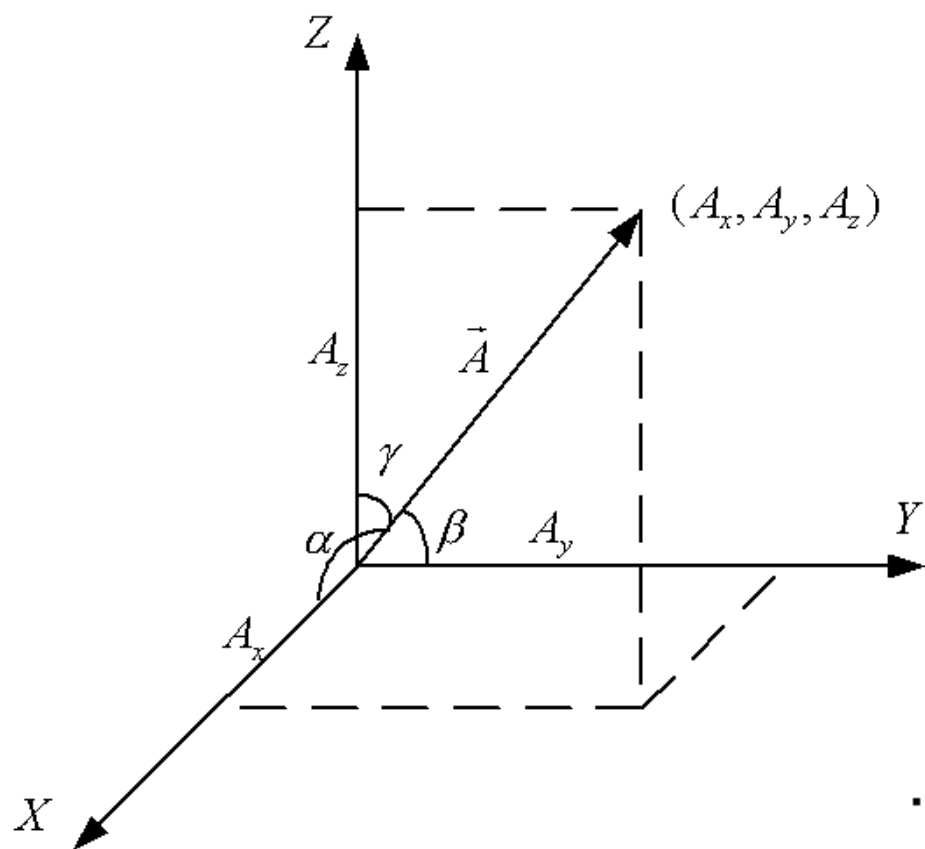
矢量数学表示： $\mathbf{A}$, $\mathbf{B}$ $\vec{A}$ $\vec{B}$

$$\vec{A} = A\hat{A} = |\vec{A}|\hat{A}$$

矢量的大小或模 A $|\vec{A}|$

单位矢量 $\hat{A}$ $|\hat{A}| = 1$

矢量用坐标分量表示



$$\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$$

$$A_x = A \cos \alpha$$

$$A_y = A \cos \beta$$

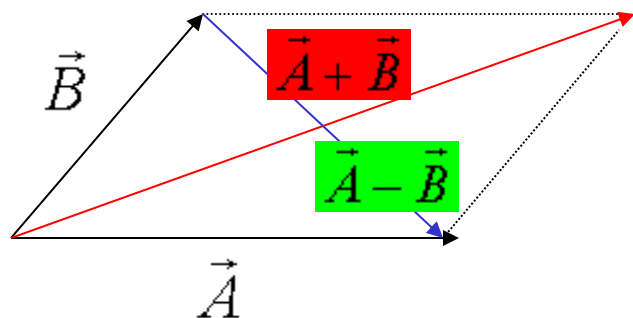
$$A_z = A \cos \gamma$$

$$\begin{aligned} \vec{A} &= A(\hat{x} \cos \alpha + \hat{y} \cos \beta + \hat{z} \cos \gamma) \\ &= A\hat{A} \end{aligned}$$

$$\therefore \hat{A} = \hat{x} \cos \alpha + \hat{y} \cos \beta + \hat{z} \cos \gamma$$

3. 矢量的代数运算

(1) 矢量的加减法



两矢量的加减在几何上是以这两矢量为邻边的平行四边形的对角线

$$\vec{A} \pm \vec{B} = (A_x \pm B_x)\hat{x} + (A_y \pm B_y)\hat{y} + (A_z \pm B_z)\hat{z}$$

矢量的加减符合交换律和结合律

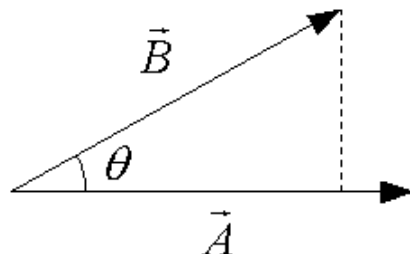
交换律: $\vec{A} + \vec{B} = \vec{B} + \vec{A}$

结合律: $\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$

(2) 标量乘矢量

$$\eta \vec{A} = \eta A_x \hat{x} + \eta A_y \hat{y} + \eta A_z \hat{z}$$

(3) 矢量的标积 (点积)



！两个矢量的标积
是一个标量

$$\vec{A} \cdot \vec{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A} \quad \text{矢量的标积符合交换律}$$

$$\vec{A} \perp \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = 0 \quad \vec{A} \parallel \vec{B} \Rightarrow \vec{A} \cdot \vec{B} = AB$$

$$\vec{A} = \vec{B} \Rightarrow A = \sqrt{A^2} = \sqrt{\vec{A} \cdot \vec{A}} = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$|\hat{A}| = \sqrt{\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma} = 1$$

特例：

$$\hat{x} \cdot \hat{x} = \hat{y} \cdot \hat{y} = \hat{z} \cdot \hat{z} = 1$$

$$\hat{x} \cdot \hat{y} = \hat{y} \cdot \hat{z} = \hat{z} \cdot \hat{x} = 0$$

$$\vec{A} \cdot \hat{x} = A_x$$

$$\vec{A} \cdot \hat{y} = A_y$$

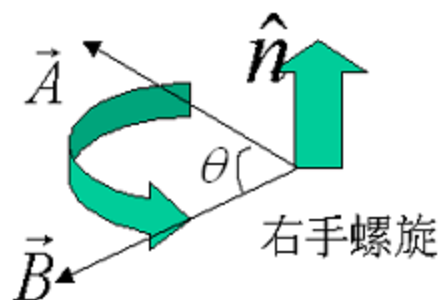
$$\vec{A} \cdot \hat{z} = A_z$$

(4) 矢量的矢积 (叉积)

$$\vec{A} \times \vec{B} = \hat{n} AB \sin \theta$$

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= (A_y B_z - A_z B_y) \hat{x} + (A_z B_x - A_x B_z) \hat{y} + (A_x B_y - A_y B_x) \hat{z}$$



$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A} \quad \text{不满足交换律}$$

性质:

$$\hat{x} \times \hat{x} = \hat{y} \times \hat{y} = \hat{z} \times \hat{z} = 0$$

$$\hat{x} \times \hat{y} = \hat{z}$$

$$\hat{y} \times \hat{z} = \hat{x}$$

$$\hat{z} \times \hat{x} = \hat{y}$$

$$\vec{A} \perp \vec{B} \Rightarrow |\vec{A} \times \vec{B}| = AB$$

$$\vec{A} \parallel \vec{B} \Rightarrow |\vec{A} \times \vec{B}| = 0$$

(5) 矢量的混合运算

$$(\vec{A} + \vec{B}) \cdot \vec{C} = \vec{A} \cdot \vec{C} + \vec{B} \cdot \vec{C}$$

$$(\vec{A} + \vec{B}) \times \vec{C} = \vec{A} \times \vec{C} + \vec{B} \times \vec{C}$$

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B})$$

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$$

1.2 三种常用坐标系中的矢量场

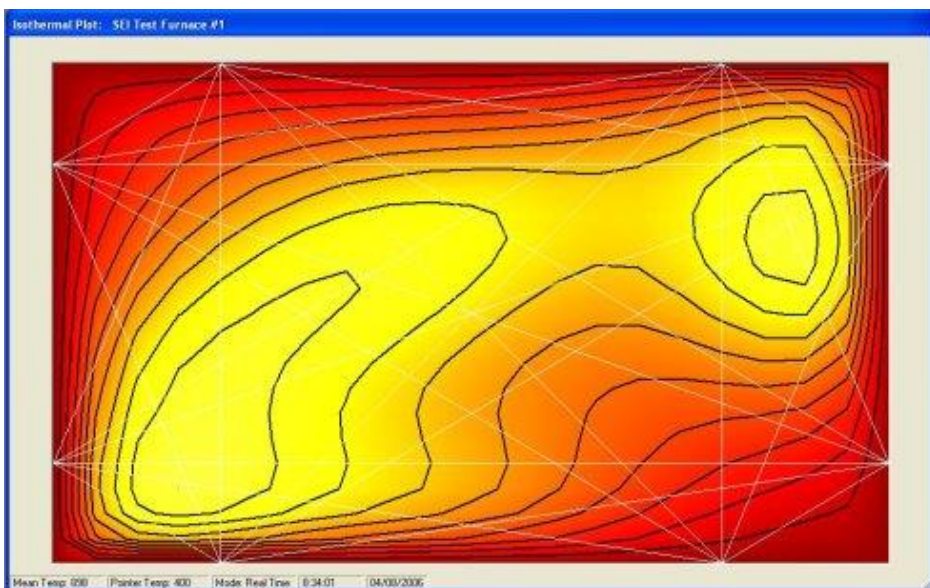
1. 场的定义：物理量在空间的分布。场是空间位置和时间
的函数，可分为标量场和矢量场。

$$H(x, y, z, t)$$

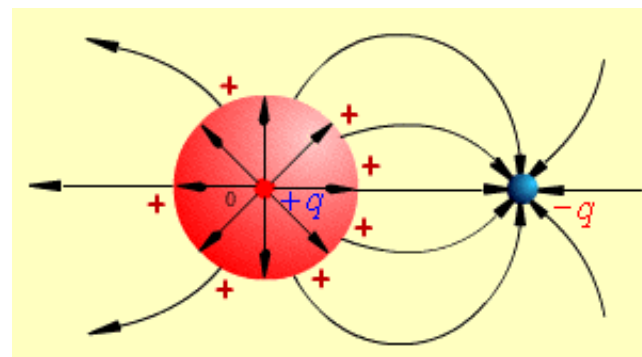
标量场可用等值线描述;

$$\vec{E}(x, y, z, t)$$

矢量场可用场线描述



温度场的等值线图



矢量场的场线

场的分类:

{ **标量场** 如温度场、电位等
矢量场 如重力场、流速

{ 静态场: 不随时间变化, 仅是空间坐标的函数, 不是时间的函数; $T(x, y, z); \vec{v}(x, y, z)$
时变场: 随时间变化, 既是空间坐标的函数也是时间的函数; $T(x, y, z, t); \vec{v}(x, y, z, t)$

场源:

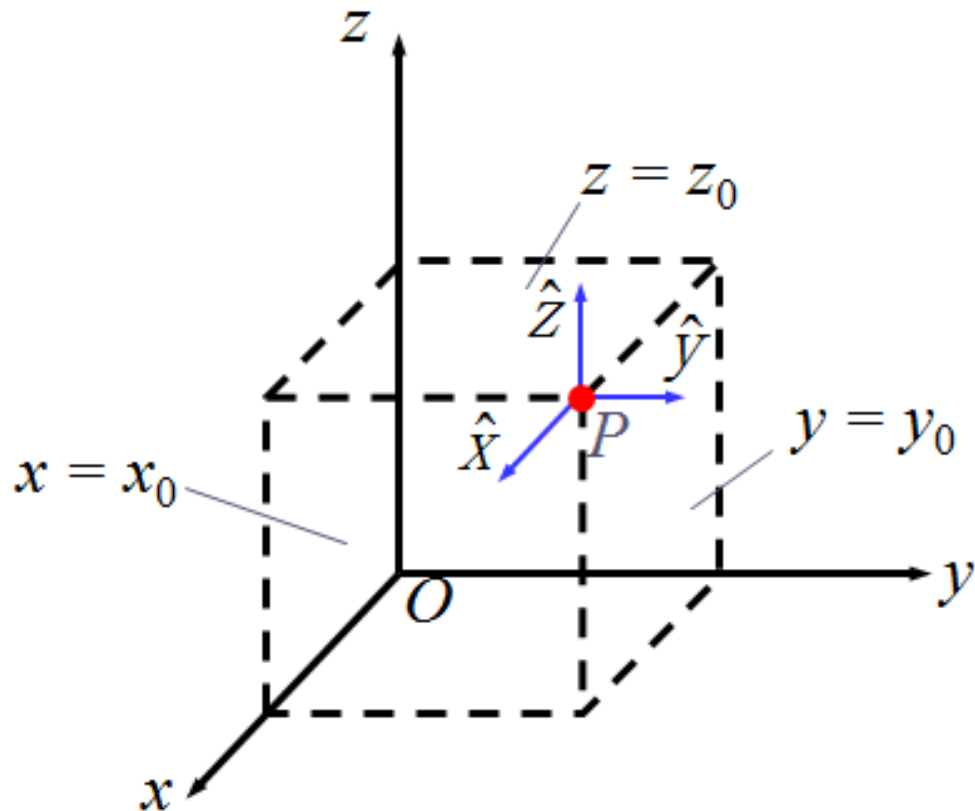
产生场的源, 场源在空间的分布用场源密度表示

场源与场的关系是因与果的关系

场源与场的关系用数学方程 (微分方程或积分方程) 描述
场的分布不仅取决于场源, 还与周围环境有关

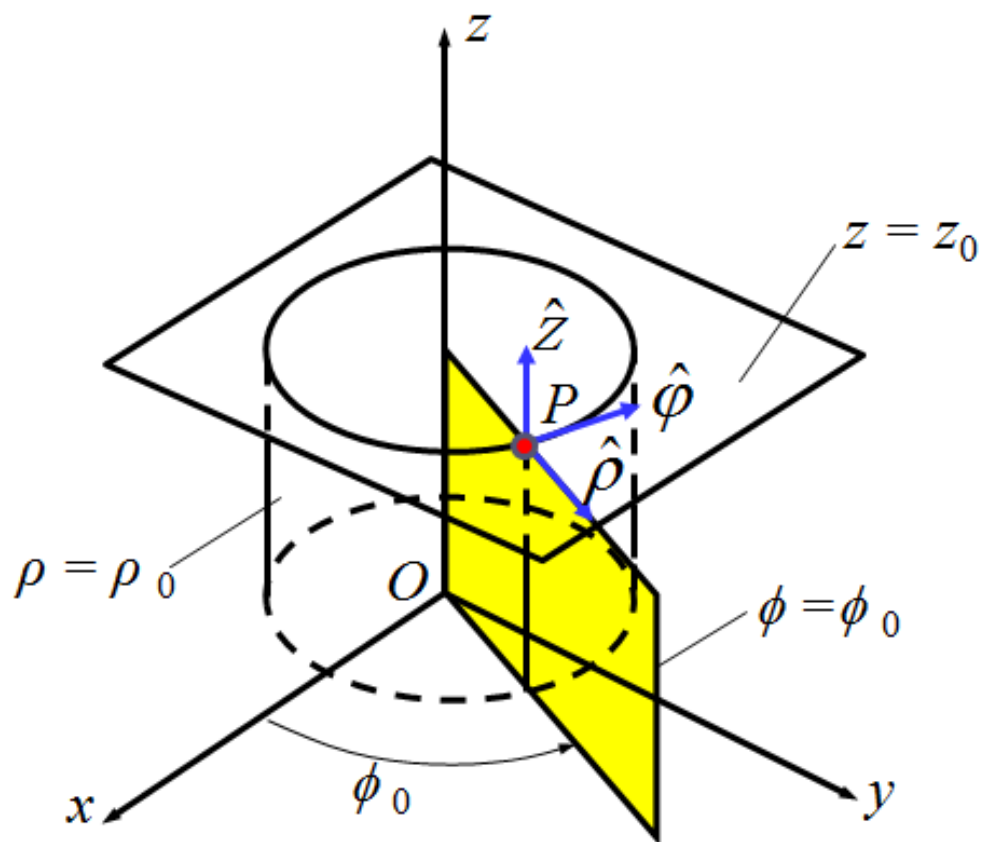
2. 空间位置的数学表述:

直角坐标系 场点P的坐标位置 (x, y, z)



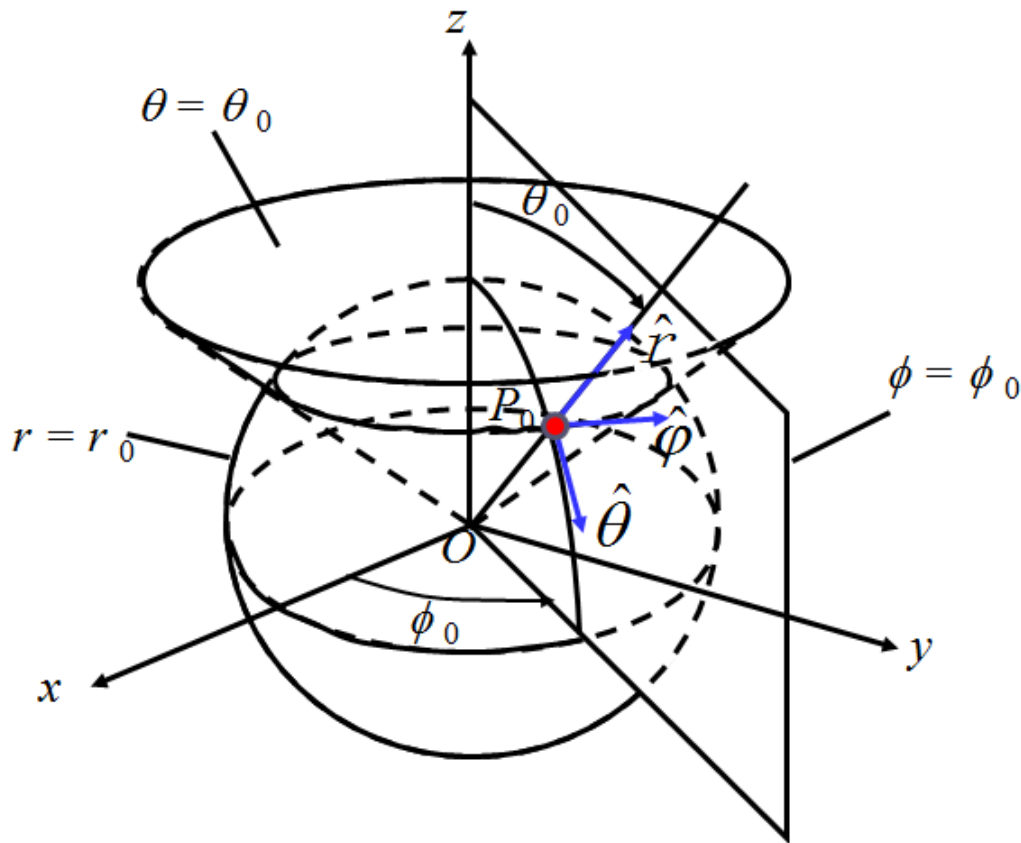
2. 空间位置的数学表述:

圆柱坐标系场点P的坐标位置 (ρ, φ, z)

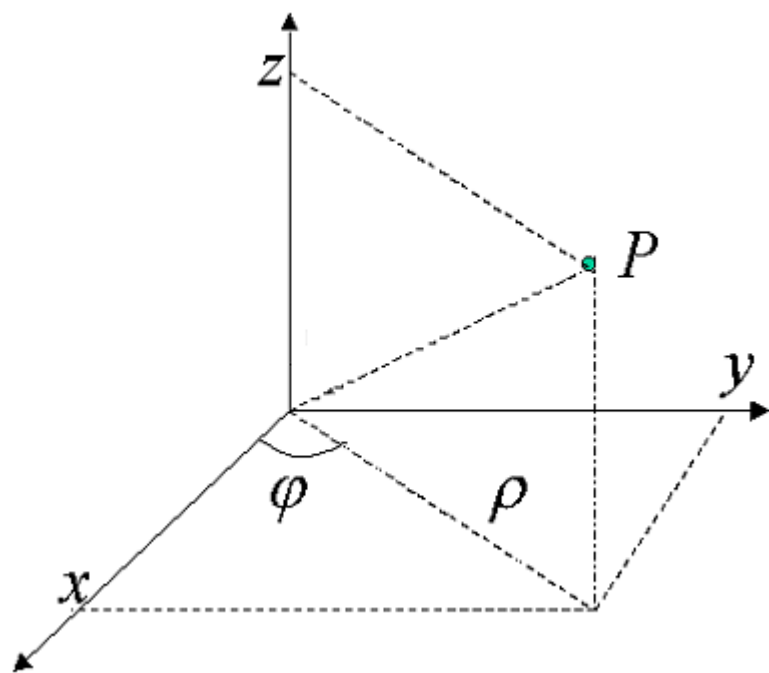


2. 空间位置的数学表述:

圆球坐标系 场点P的坐标位置 (r, θ, φ)



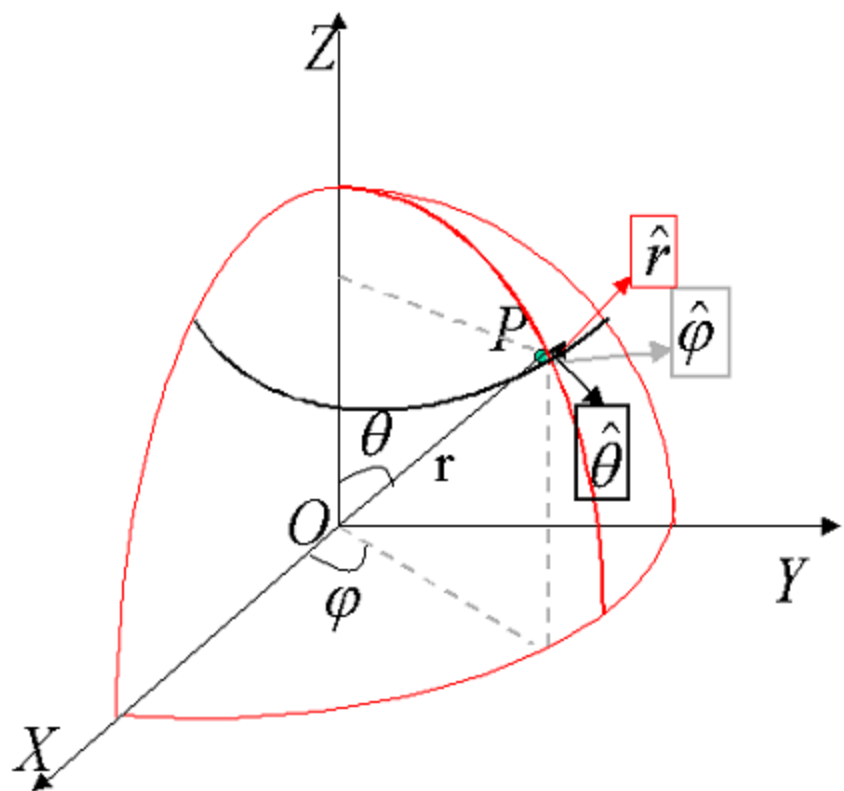
直角坐标系坐标与圆柱坐标系坐标的关系



$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \\ z = z \end{cases}$$

$$\begin{cases} \rho = \sqrt{x^2 + y^2} \\ \varphi = \arctan \frac{y}{x} \\ z = z \end{cases}$$

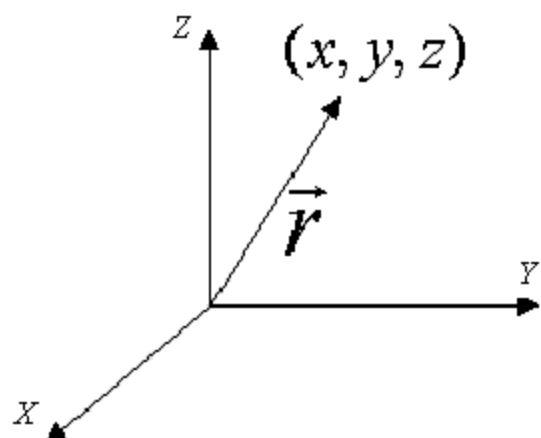
直角坐标系坐标与圆球坐标系坐标的关系



$$\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$$

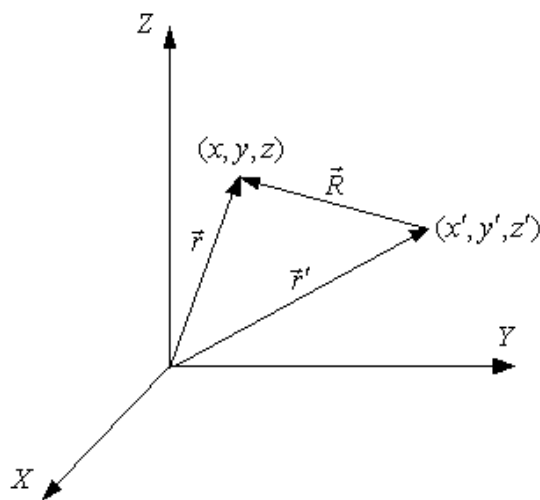
$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \arctan \frac{\sqrt{x^2 + y^2}}{z} \\ \varphi = \arctan \frac{y}{x} \end{cases}$$

位置矢量



$$\begin{aligned}(x, y, z) & \quad \vec{r} = x\hat{x} + y\hat{y} + z\hat{z} \\ (\rho, \varphi, z) & \Rightarrow \quad = \rho\hat{\rho} + \varphi\hat{\varphi} + z\hat{z} \\ (r, \theta, \varphi) & \quad = r\hat{r} + \theta\hat{\theta} + \varphi\hat{\varphi} \\ f(x, y, z) & \\ f(\rho, \varphi, z) & \Rightarrow f(\vec{r}) \\ f(r, \theta, \varphi) & \end{aligned}$$

距离矢量



$$\vec{R} = \vec{r} - \vec{r}' = (x - x')\hat{x} + (y - y')\hat{y} + (z - z')\hat{z}$$

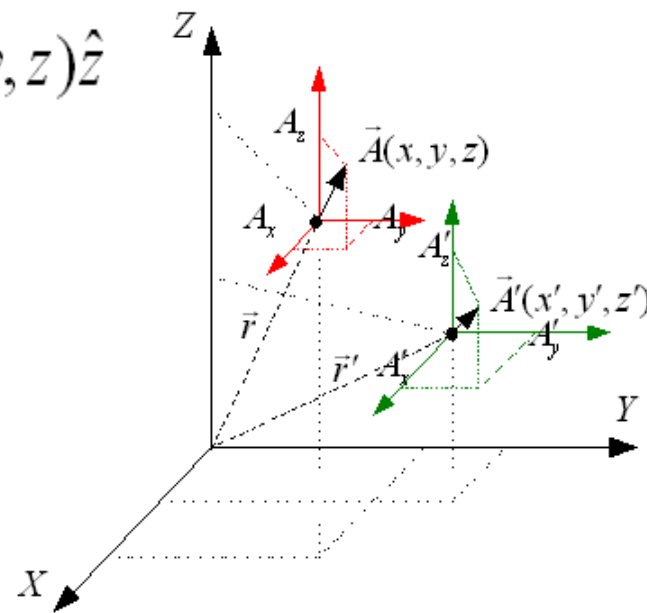
$$R = |\vec{r} - \vec{r}'| = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2}$$

3. 矢量场的表示:

(1) 直角坐标系中矢量场的表示

$$\begin{aligned}\vec{A}(x, y, z) &= A_x(x, y, z)\hat{x} + A_y(x, y, z)\hat{y} + A_z(x, y, z)\hat{z} \\ &= A_x(\vec{r})\hat{x} + A_y(\vec{r})\hat{y} + A_z(\vec{r})\hat{z}\end{aligned}$$

$$\begin{aligned}\vec{A}'(x', y', z') &= A'_x(x', y', z')\hat{x} + A'_y(x', y', z')\hat{y} \\ &\quad + A'_z(x', y', z')\hat{z} \\ &= A'_x(\vec{r}')\hat{x} + A'_y(\vec{r}')\hat{y} + A'_z(\vec{r}')\hat{z}\end{aligned}$$



- 位置矢量的坐标表示与矢量场值的表示相互独立。矢量场沿三个坐标轴的分量又是空间位置的函数。

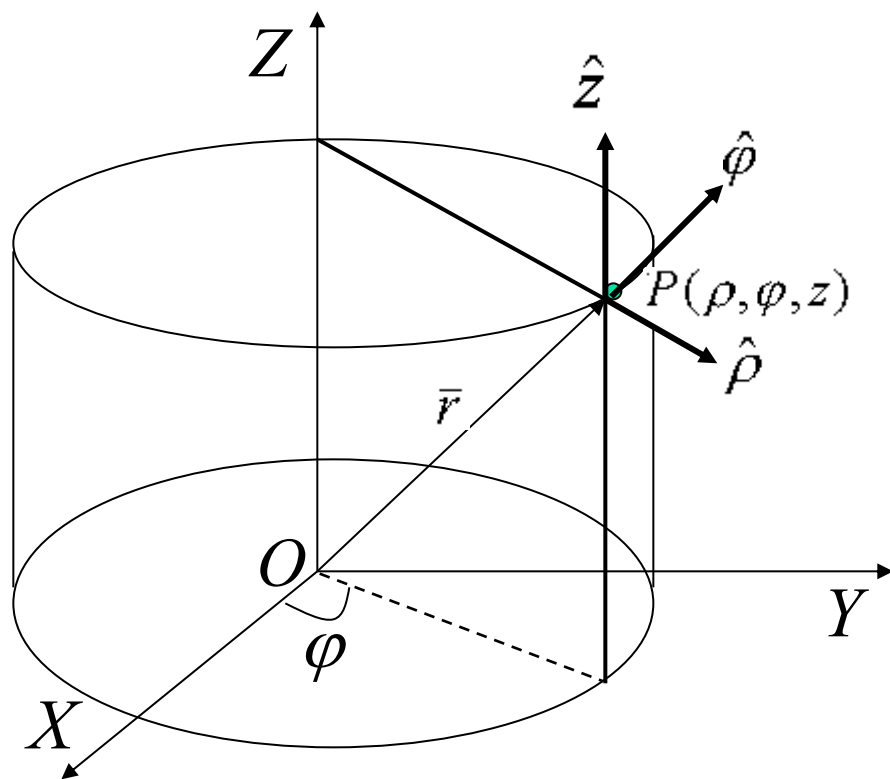
(2) 圆柱坐标系中矢量场的表示

- 圆柱坐标轴单位矢量 $\hat{\rho}$ $\hat{\phi}$ $\hat{z}$

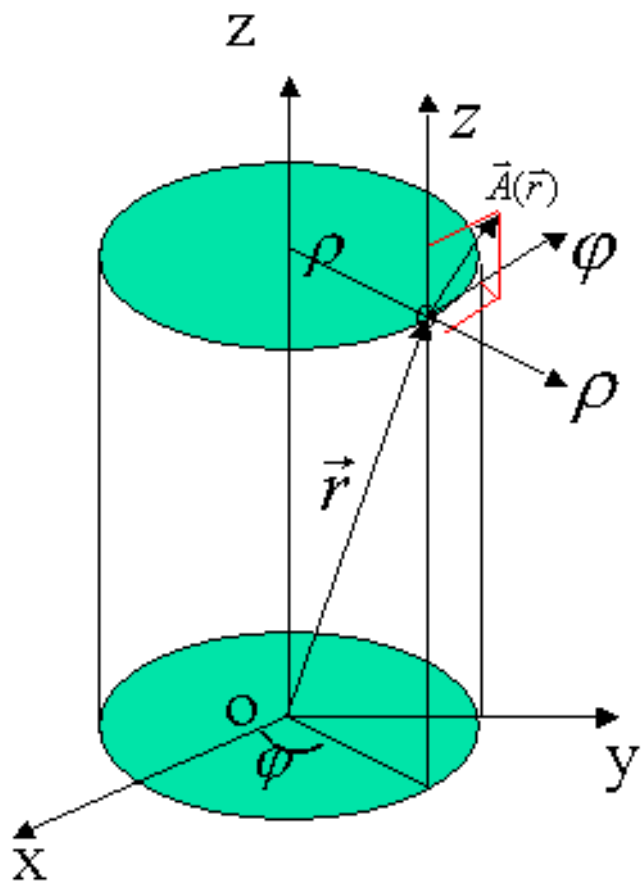
- 直角坐标系和圆柱单位矢量的关系：

$$\begin{cases} \hat{\rho} = \hat{x} \cos \varphi + \hat{y} \sin \varphi \\ \hat{\phi} = -\hat{x} \sin \varphi + \hat{y} \cos \varphi \end{cases}$$

$$\begin{cases} \hat{x} = \hat{\rho} \cos \varphi - \hat{\phi} \sin \varphi \\ \hat{y} = \hat{\rho} \sin \varphi + \hat{\phi} \cos \varphi \end{cases}$$



矢量场的圆柱坐标系分量



$$\vec{A}(\vec{r}) = A_{\rho}(\vec{r})\hat{\rho} + A_{\phi}(\vec{r})\hat{\phi} + A_z(\vec{r})\hat{z}$$

$$\begin{cases} A_{\rho}(\vec{r}) = \vec{A}(\vec{r}) \cdot \hat{\rho} \\ A_{\phi}(\vec{r}) = \vec{A}(\vec{r}) \cdot \hat{\phi} \\ A_z(\vec{r}) = \vec{A}(\vec{r}) \cdot \hat{z} \end{cases}$$

矢量直角坐标分量与柱坐标分量的转换矩阵：

$$\begin{bmatrix} A_\rho \\ A_\varphi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\varphi \\ A_z \end{bmatrix}$$

例：将 $\vec{A}(\vec{r}) = x\hat{x} + y\hat{y}$; $\vec{B}(\vec{r}) = x\hat{y} - y\hat{x}$ 用圆柱坐标分量表示

解：

$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \\ z = z \end{cases} \quad \begin{cases} \hat{x} = \hat{\rho} \cos \varphi - \hat{\varphi} \sin \varphi \\ \hat{y} = \hat{\rho} \sin \varphi + \hat{\varphi} \cos \varphi \end{cases}$$

$$\begin{aligned} \vec{A}(\vec{r}) &= x\hat{x} + y\hat{y} \\ &= \rho \cos \varphi (\hat{\rho} \cos \varphi - \hat{\varphi} \sin \varphi) + \rho \sin \varphi (\hat{\rho} \sin \varphi + \hat{\varphi} \cos \varphi) \\ &= \rho \hat{\rho} \end{aligned}$$

$$\begin{aligned} \vec{B}(\vec{r}) &= x\hat{y} - y\hat{x} \\ &= \rho \cos \varphi (\hat{\rho} \sin \varphi + \hat{\varphi} \cos \varphi) - \rho \sin \varphi (\hat{\rho} \cos \varphi - \hat{\varphi} \sin \varphi) \\ &= \rho \hat{\varphi} \end{aligned}$$

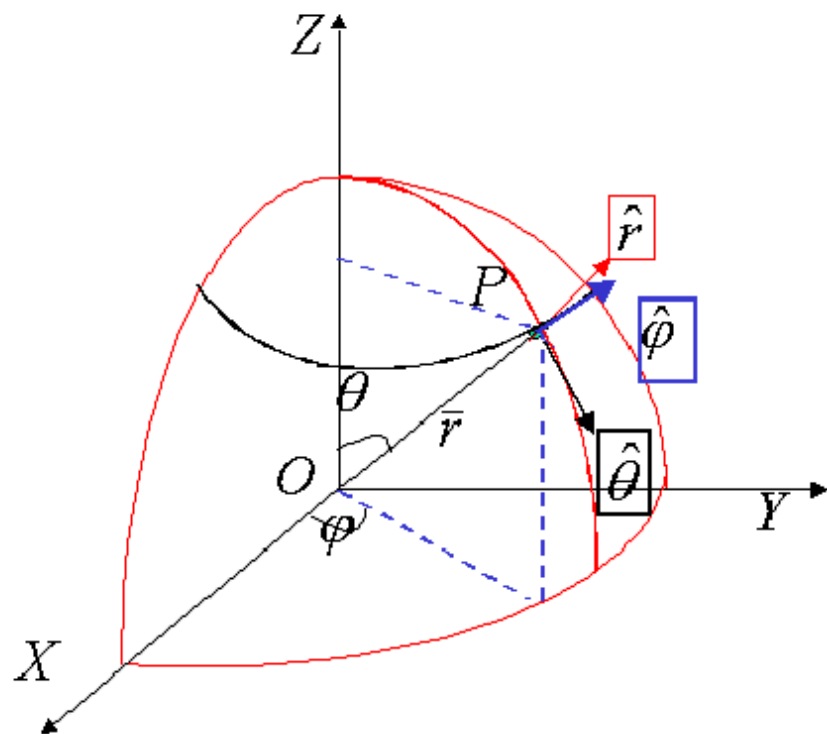
特点：当矢量场的方向为圆柱面的法向或切向时，用圆柱坐标表示不但形式简单，而且形象，更易理解。

(3) 矢量场的圆球坐标系分量

- 圆柱坐标轴单位矢量 $\hat{r}$ $\hat{\theta}$ $\hat{\phi}$
- 圆球坐标系与直角坐标系单位矢量的关系：

$$\begin{cases} \hat{r} = \hat{x} \sin \theta \cos \varphi + \hat{y} \sin \theta \sin \varphi + \hat{z} \cos \theta \\ \hat{\theta} = \hat{x} \cos \theta \cos \varphi + \hat{y} \cos \theta \sin \varphi - \hat{z} \sin \theta \\ \hat{\phi} = \hat{x} \sin \varphi + \hat{y} \cos \varphi \end{cases}$$

$$\begin{cases} \hat{x} = \hat{r} \sin \theta \cos \varphi + \hat{\theta} \cos \theta \cos \varphi - \hat{\phi} \sin \varphi \\ \hat{y} = \hat{r} \sin \theta \sin \varphi + \hat{\theta} \cos \theta \sin \varphi + \hat{\phi} \cos \varphi \\ \hat{z} = \hat{r} \cos \theta - \hat{\theta} \sin \theta \end{cases}$$



- 矢量场的圆球坐标系分量

$$\vec{A}(r, \theta, \varphi) = A_r(r, \theta, \varphi)\hat{r} + A_\theta(r, \theta, \varphi)\hat{\theta} + A_\varphi(r, \theta, \varphi)\hat{\varphi}$$

- 不同的位置上，圆球坐标轴单位矢量方向不同；
- 当矢量场的方向为圆球面的法向或切向时，用圆球坐标表示不但形式简单，而且形象，更易理解。

- 矢量场直角坐标分量与球坐标分量的转换矩阵：

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\varphi \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\varphi & \sin\theta\sin\varphi & \cos\theta \\ \cos\theta\cos\varphi & \cos\theta\sin\varphi & -\sin\theta \\ -\sin\varphi & \cos\varphi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\varphi & \cos\theta\cos\varphi & -\sin\varphi \\ \sin\theta\sin\varphi & \cos\theta\sin\varphi & \cos\varphi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\varphi \end{bmatrix}$$